

Frank Kusi Appiah

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Oberlin, OH

Technical Program Management · Systems Builder · AI / IoT

EDUCATION

Oberlin College — College of Arts & Sciences Aug 2024 – May 2027
B.A., 3-2 Engineering Program (B.A. + B.S. Engineering) | GPA: 3.52
Expected B.A. May 2027

Columbia University — Fu Foundation School of Engineering and Applied Science 2027 – 2029
3-2 Engineering continuation; expected engineering degree 2029
Coursework: Data Structures · Discrete Math · Multivariable Calculus · Engineering Physics · Mechanics & Relativity · Intro CS · Principles of Economics
Honors: STRONG Scholar · Arthur Blank Fellow · MLK Internship Program Recipient · Oberlin AI Micro Grant Recipient

TECHNICAL SKILLS

Program & Systems: Agile/Scrum, technical requirements, system design, failure-mode analysis, TDD, code review, documentation, data pipeline design, RESTful API design, RBAC
Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL, Swift
Platforms & Tools: AWS (Cognito, S3, Lambda, RDS), Firebase, GitHub, CI/CD, Linux/Unix, SolarWinds, Grafana, MATLAB, Postman, VS Code, Fusion 360
Frameworks & Data: Node.js, React, Tailwind CSS, Flutter, NumPy, OpenCV, PostgreSQL, MySQL, Firebase Realtime DB, GraphQL

EXPERIENCE

Oberlin Environmental Dashboard Jun 2025 – Present
Hardware & Software Manager

- Owned the full infrastructure lifecycle—architecture, deployment, testing, and documentation—for a system serving the entire Oberlin campus.
- Engineered Python and SQL pipelines ingesting, validating, and storing 10,000+ sensor readings/day across 30+ IoT nodes, with zero data loss since deployment.
- Shipped REST APIs and real-time dashboards tracking uptime, latency, and data integrity across 25+ deployments, creating network-wide visibility for operational decisions.
- Cut system failure rates by 40% through automated anomaly detection that flags out-of-range and inconsistent readings before production.
- Designed solar-powered charging systems with intelligent battery management that maintained 99.5% uptime through all weather conditions.

Anansenet Limited Nov 2024 – Jan 2025
Software Engineer Intern

- Secured connected-hardware infrastructure with AWS Cognito authentication and RBAC, cutting unauthorized access incidents by 35%.
- Reviewed 50+ pull requests for edge-case coverage, failure-mode analysis, and test reliability, improving release stability before production deployments.
- Shipped production TypeScript and Tailwind CSS interfaces used by real customers, with zero reported layout regressions post-launch.

Oberlin College — Center for Information Technology Aug 2024 – Dec 2025
Help Desk Technology Consultant

- Resolved 40+ software, hardware, and network issues per week by reproducing problems systematically and validating fixes before closure.
- Built a knowledge base of 300+ documented incidents in SolarWinds, surfacing recurring failure patterns and enabling faster team-wide resolution.
- Wrote troubleshooting guides adopted across the team that reduced resolution time for common hardware and software issues.

SELECTED PROJECTS & LEADERSHIP

PDF Reader for Even G2

[Personal Project](#) · [Wearable Computing](#) · [PDF/DOCX Rendering](#)

- Launched a wearable smart-glasses document reader on Even Hub, reaching 1,600+ users; owned product requirements, testing, and release readiness.
- Identified that the device's native text renderer could not reliably preserve symbols, custom fonts, tables, or page layouts; designed an image-first pipeline that renders PDF/DOCX pages as high-resolution visual panels.
- Implemented Smart Flow panel segmentation and resilient tiled display delivery, preventing awkward content splits and enabling users to resume reading after reconnecting.

AI Community Calendar Aggregator

[Oberlin AI Micro Grant](#) · [Multi-agent AI](#) · [Human-in-the-loop](#)

- Built a grant-funded multi-agent system that discovers and extracts event data from 10+ institutions, replacing a fully manual process.
- Designed an LLM extraction pipeline and human-in-the-loop review workflow that feeds a live calendar on 23+ public display signs and a weekly newsletter.
- Selected as a pilot for expansion to MidTown Cleveland and the broader Cleveland Neighborhood Progress network, validating scalability beyond Oberlin.

OberiOS — Community Intelligence Platform

[Personal Project](#) · [Node.js](#) · [PostgreSQL](#) · [AWS](#) · [ML ranking](#)

- Architected a platform combining live IoT data, AI-aggregated events, and a jobs board into one personalized feed to solve Oberlin's fragmented information problem.
- Built REST and GraphQL services, normalized data from different sources, deployed on AWS, and established a GitHub Actions CI/CD pipeline.

Leadership: Oberlin College Soccer Club Captain (operations for 22+ players) · Oberlin Robotics Club · Oberlin AI Advisory Group · Obie Leaders Leadership Development Series